



Zwuwur

Lateral Transfer Marketplace for Bank Employees

Software Requirements Specification (SRS)

Version 1.0

Date: July 21, 2026

*Stack: Express.js (Node.js) · MySQL 8 · Redis · BullMQ · Telegram Bot & Mini App · Admin PWA · Pure JavaScript
(raw SQL via mysql2/promise) · Local file storage (multer)*

1. Introduction

1.1 Purpose

This document specifies the requirements for Zwuwur, a lateral-transfer information marketplace for bank employees in Ethiopia. An employee registers their current work location (region, zone, branch, neighborhood) and grade, and optionally declares one or more desired transfer destinations. The system matches employees within the same bank and surfaces them in a ranked marketplace feed. A candidate's contact information is revealed only after the viewing employee completes a paid unlock.

Zwuwur does not manage, approve, or execute the transfer itself — that process happens directly between the two employees and their institution, outside the system.

The product is delivered through:

- Telegram Bot — primary entry point; minimal conversational onboarding and all notifications.
- Telegram Mini App — rich in-Telegram UI; profile completion, marketplace (Feed + People), purchases, and interest management. Four tabs.
- Admin PWA — staff-facing monitoring and reference-data console.

Supported Languages: English and Amharic — across the entire product (UI, reference data, notifications, and user-entered names). Every piece of text a human reads carries both an English and an Amharic version.

1.2 Objectives

The Release aims to:

- Digitize discovery of lateral transfer opportunities among bank employees.
- Provide low-friction, two-phase registration (fast bot onboarding + richer Mini App profile completion).
- Provide accurate zone-level matching, strictly scoped within a single bank, with progressive broadening so the feed never empties.
- Monetize contact visibility through a pay-per-reveal marketplace.
- Give administrators tools to manage reference data, monitor activity, and broadcast notifications.
- Build a maintainable, scalable foundation (1,000+ users from day one) using a simple, flat data model.

1.3 Intended Audience

This document is intended for:

- Business Stakeholders
- Project Sponsors
- Product Owners
- Business Analysts
- Software Developers

- QA / Test Engineers
- System Administrators

2. Scope

2.1 Modules in this Release

- User Registration & Authentication (Telegram-based, two-phase)
- Profile Completion & Management (Mini App)
- Profile Completeness Gating & Nudging
- Photo Management (Telegram auto-fetch + optional local upload)
- Reference Data Management (banks, regions, zones, grades — admin-seeded)
- Transfer Interest Management (region/zone multi-select with constraints)
- Marketplace Feed (ranked, endless scroll, progressive broadening)
- People Tab (candidates in the user's desired areas)
- Pay-to-Reveal Purchases
- Payment Processing (provider-agnostic: Chapa and/or Telebirr)
- Notification Management (digest, payment confirmation, broadcast, profile nudge)
- Admin Monitoring, Reporting & Broadcast
- Staff Management & Role-Based Access Control
- Telegram Bot Application
- Telegram Mini App (4 tabs)
- Admin PWA

2.2 Out of Scope for this Release

- Management or approval of the actual physical transfer/relocation process.
- In-app messaging or negotiation between matched parties (contact is revealed; the conversation happens off-platform).
- Institutions other than banks (planned future enhancement).
- Multi-party chain matching ($A \rightarrow B \rightarrow C \rightarrow A$). This release supports pairwise lead discovery only.
- PDF/Excel report export (on-screen reports only).
- Email/SMS notification fallback (Telegram-only).
- Automated KYC / identity verification.

3. Stakeholders

3.1 External Users

- Registered Bank Employees — marketplace participants (both viewers and candidates).

3.2 Internal Users

- Super Administrator — full admin control including staff management.
- Administrator — reference data, user monitoring, broadcasts, reports.

4. Business Workflow

Phase 1 — Bot Onboarding (fast, ~30 seconds)

- User starts the Telegram bot (/start).
- User selects language (English / Amharic).
- User shares their phone number (Telegram native contact-share; OTP stub as fallback).
- User selects their Bank (31 seeded banks).
- User selects their Region (14 seeded regions/chartered cities).
- User selects their Zone (zones under the selected region; subcities for Addis Ababa).
- System creates the user with status registered (incomplete) and prompts the user to open the Mini App to finish setup.

Phase 2 — Mini App Profile Completion (rich, validated)

- User opens the Mini App and is shown a “Complete your profile” gate.
- User enters Full Name (at least one language required) and optionally the other language version.
- User selects Grade (band → grade number).
- User enters Branch Name (at least one language required).
- User enters Neighborhood (at least one language required).
- System auto-fetches the user's Telegram profile photo as the default avatar (user may later upload a custom one).
- On saving all required fields, the profile becomes complete and marketplace features unlock.
- User is prompted (skippable) to add Transfer Interests.

Phase 3 — Transfer Interest Submission

- In the Mini App, the user selects up to 3 regions, and up to 3 zones per region.
- Selecting a zone auto-selects its parent region; a region may be selected without any zone (a “broad” interest = anywhere in that region).
- The user may save at any time (even a single zone) and edit interests later.

Phase 4 — Marketplace Feed Generation

- User opens the Feed tab (endless scroll).
- The system runs a live query for all other complete, active users of the same bank, within ± 1 grade band, who have expressed interest in the user's zone (zone-level) or region (region-broad).

- Results are ranked: mutual matches first, then by match specificity (zone above region), then grade proximity, then recency.
- As the user scrolls, zone-level matches exhaust first and region-broad matches appear — the feed never empties while anyone in the bank wants the user's region.
- Feed cards show public info; paywalled fields appear as * until purchased.

Phase 5 — People Tab

- User opens the People tab to see candidates already located in their desired areas.
- Requires interests to be set; otherwise an empty state prompts the user to add interests.

Phase 6 — Purchase & Reveal

- User taps “Unlock Contact” on a card and confirms the price.
- System creates a purchase + pending payment and returns a checkout URL (Chapa or Telebirr hosted page).
- User pays off-platform; the provider's webhook confirms payment (HMAC-verified, idempotent).
- The candidate's contact details are revealed to the buyer, in both the Feed and the Purchases tab. The purchase is permanent and never charged twice.

Phase 7 — Notification

- A daily digest notifies users of new qualifying interests in their area.
- Payment confirmations and admin broadcasts are delivered via Telegram.
- Incomplete profiles receive a “complete your profile” nudge.

Phase 8 — Post-Reveal (Outside the Platform)

- The two employees communicate and negotiate the transfer directly. Zwuwur does not track, mediate, or approve the outcome.

5. Functional Requirements

5.1 Registration & Authentication

FR-AUTH-001 — Users shall register through the Telegram Bot using their Telegram account.

FR-AUTH-002 — Users shall verify their phone number via Telegram contact-share (OTP stub as fallback).

FR-AUTH-003 — The system shall prevent duplicate accounts by Telegram ID and by (phone number, bank) pair.

FR-AUTH-004 — Each user account shall be associated with exactly one bank, fixed at registration.

FR-AUTH-005 — Mini App requests shall be authenticated by validating Telegram WebApp initData signatures.

FR-AUTH-006 — Admin PWA users shall authenticate with email + password.

FR-AUTH-007 — Admin passwords shall be stored using secure hashing (bcrypt).

FR-AUTH-008 — Bot onboarding shall collect only: language, phone, bank, region, zone. All other profile fields are collected in the Mini App.

5.2 Profile Completion & Management

FR-PROFILE-001 — The Mini App shall collect Full Name, Grade, Branch Name, and Neighborhood.

FR-PROFILE-002 — Full Name, Branch Name, and Neighborhood shall each be bilingual fields (_en and _am), with at least one language version required per field. Neither pair may be entirely empty.

FR-PROFILE-003 — Branch Name shall require a minimum of 3 characters in the filled language version.

FR-PROFILE-004 — Neighborhood shall require a minimum of 2 characters in the filled language version.

FR-PROFILE-005 — Grade shall be selected via a band-then-number picker (bands of 3: 1-3, 4-6, ..., 16-18).

FR-PROFILE-006 — Users shall be able to view and update their profile (branch, neighborhood, grade, names, photo, language, region, zone) in the Mini App. Region and zone changes are fully self-service (no admin verification required), as the data is pre-seeded and validated against the reference tables.

FR-PROFILE-007 — Bank shall not be changeable after registration.

5.3 Profile Completeness Gating & Nudging

FR-COMP-001 — A profile shall be complete only when, for each of {Full Name, Branch Name, Neighborhood}, at least one language version is filled, AND Grade is set.

FR-COMP-002 — Marketplace features (Feed, People, Purchases, appearing in others' feeds) shall be gated behind a complete profile.

FR-COMP-003 — The Mini App shall show a “Complete your profile” gate screen to incomplete users.

FR-COMP-004 — The system shall compute a profile-completeness state covering encouraged fields: the missing language version of each bilingual pair, ≥ 1 transfer interest, and custom photo.

FR-COMP-005 — A completeness nudge banner shall appear at the top of the Feed and in the Profile tab until the profile is 100% complete. The nudge shall be language-aware: if the user filled Amharic only, it prompts for English (and vice versa).

FR-COMP-006 — The bot shall send a one-time “finish setting up” reminder with a Mini App deep-link after bot onboarding completes.

5.4 Photo Management

FR-PHOTO-001 — On profile creation, the system shall fetch the user's Telegram profile photo via the Bot API (getUserProfilePhotos) and store it locally as the default avatar.

FR-PHOTO-002 — If the user has no Telegram profile photo, the card shall display a placeholder avatar.

FR-PHOTO-003 — Users may optionally upload a custom photo in the Mini App, which overrides the Telegram photo.

FR-PHOTO-004 — Photo uploads shall be stored on local disk (multer diskStorage); no cloud object storage (AWS/S3/R2) shall be used.

FR-PHOTO-005 — Uploads shall be validated: max 5 MB; MIME whitelist (image/jpeg, image/png, image/webp); sanitized unique filenames to prevent path traversal.

FR-PHOTO-006 — Users may remove a custom photo (reverting to the Telegram photo or placeholder).

FR-PHOTO-007 — The photo is a public card field (visible to all same-bank users).

5.5 Reference Data Management (Admin-Seeded)

FR-REF-001 — Administrators shall create and maintain the list of participating banks (English + Amharic names, nickname).

FR-REF-002 — Administrators shall create and maintain regions (14 seeded) and zones (111 seeded) as two flat tables; zones reference their parent region by foreign key.

FR-REF-003 — For Addis Ababa, zones shall represent subcities.

FR-REF-004 — Administrators shall create and maintain the grade matrix (18 grades, grouped into bands of 3), with English + Amharic labels.

FR-REF-005 — Reference data shall not be deletable while active users or interests reference it; deactivation shall be blocked in that case.

FR-REF-006 — All reference-data name fields shall have mandatory English and Amharic columns (NOT NULL).

5.6 Transfer Interest Management

FR-INT-001 — Users shall declare transfer interests in the Mini App as region and/or zone selections.

FR-INT-002 — A user may select a maximum of 3 regions total.

FR-INT-003 — A user may select a maximum of 3 zones per selected region.

FR-INT-004 — Selecting a zone shall automatically count its parent region as selected.

FR-INT-005 — A region may be selected without any zone, representing a broad interest ("anywhere in this region").

FR-INT-006 — A selected zone shall always be stored with its parent region.

FR-INT-007 — Users may save their selection at any time (including a single zone) and may add, edit, or remove interests later.

FR-INT-008 — The interest-selection UI shall open on the user's home region's zones by default, with a back navigation to the full region list.

FR-INT-009 — Each interest shall be timestamped at creation for ranking and digest purposes.

5.7 Marketplace Feed

FR-FEED-001 — The Feed shall be generated by a live query at request time; matches shall not be precomputed or persisted.

FR-FEED-002 — The Feed shall show candidates who have expressed interest in the viewer's zone (zone-level) or the viewer's region (region-broad interest only, where the interest's zone_id is null).

FR-FEED-003 — Matching shall be strictly scoped to the same bank; the system shall never return a cross-bank candidate.

FR-FEED-004 — Candidates shall be included only if their grade is within ± 1 band of the viewer's grade (bands of 3).

FR-FEED-005 — Only complete and active candidates shall appear; only complete viewers may browse.

FR-FEED-006 — The Feed shall be ranked: mutual match first, then match specificity (zone above region), then grade proximity (smaller grade distance first), then interest recency.

FR-FEED-007 — A match is mutual when the viewer also has an interest in the candidate's zone or region.

FR-FEED-008 — The Feed shall be an endless scroll; zone-level matches shall appear before region-broad matches so the feed broadens naturally and never empties while candidates want the viewer's region.

FR-FEED-009 — The Feed shall provide a “back to top” control after scrolling.

FR-FEED-010 — Users may refresh the feed on demand to reflect newly submitted interests.

FR-FEED-011 — Unpurchased cards shall show public fields only; paywalled fields shall be masked (see §5.9 API contract).

5.8 People Tab

FR-PEOPLE-001 — The People tab shall show candidates who are located in the viewer's desired areas (the viewer has an interest in the candidate's zone, or a broad interest in the candidate's region).

FR-PEOPLE-002 — The People tab shall require the viewer to have at least one interest; with no interests it shall show an empty state prompting the user to add interests.

FR-PEOPLE-003 — The People tab shall apply the same-bank and ± 1 -band constraints.

FR-PEOPLE-004 — The People tab shall be ranked by grade proximity, then recency.

FR-PEOPLE-005 — Cards shall follow the same public/paywalled visibility rules as the Feed, including the mutual-match (swap) badge.

5.9 Pay-to-Reveal Purchases & API Response Contract

FR-PUR-001 — Users shall be able to initiate a purchase to reveal a candidate's contact details.

FR-PUR-002 — Contact details shall be revealed only after payment is confirmed.

FR-PUR-003 — Each purchase shall be permanently recorded against the buyer/candidate pair.

FR-PUR-004 — The system shall prevent charging the same buyer twice for the same candidate.

FR-PUR-005 — Revealed contact details shall include: Full Name (all available language versions), Phone, Telegram username, Branch Name (all available language versions), and Neighborhood (all available language versions).

FR-PUR-006 — A purchased reveal shall be visible to the buyer in both the Feed and the Purchases tab.

FR-PUR-007 — API response contract for card fields: The backend shall return paywalled fields as “*” (masked) for unpurchased candidates, and as full values for purchased candidates. The frontend renders whatever the API returns without implementing client-side paywall logic.

Example — unpurchased card:

```
{
  "id": 4821,
  "bank_id": 3,
  "grade": { "band": 2, "number": 5 },
  "region": "Oromia",
  "zone": "Shashamane",
```



```

"full_name_en": "*",
"full_name_am": "*",
"branch_name_en": "*",
"branch_name_am": "*",
"neighborhood_en": "*",
"neighborhood_am": "*",
"phone_number": "*",
"telegram_username": "*",
"photo_url": "https://cdn.zwuur.app/avatars/4821.jpg",
"match_type": "zone",
"is_mutual": true,
"purchased": false
}

```

Example — purchased card (same candidate, after payment):

```

{
  "id": 4821,
  "bank_id": 3,
  "grade": { "band": 2, "number": 5 },
  "region": "Oromia",
  "zone": "Shashamane",
  "full_name_en": "Betelhem Getachew",
  "full_name_am": "በቲአህም ገታቸ",
  "branch_name_en": "Shashamane Main Branch",
  "branch_name_am": "ሸሸማኔ ዋና ቅርንጾ በት",
  "neighborhood_en": "Arada Kebele 02",
  "neighborhood_am": "አራዳ ከበሌ 02",
  "phone_number": "+251911223344",
  "telegram_username": "@betiget",
  "photo_url": "https://cdn.zwuur.app/avatars/4821.jpg",
  "match_type": "zone",
  "is_mutual": true,
  "purchased": true
}

```

Design principle: The frontend is a dumb renderer. It displays whatever the API returns. If a field is "", it shows *. If it's a real value, it shows the value. No client-side encryption, decryption, or paywall logic. The backend is the single source of truth for what is revealed.*

5.10 Payment Management

FR-PAY-001 — The system shall integrate with a payment provider via a provider-agnostic interface supporting Chapa and/or Telebirr (final provider to be confirmed).

FR-PAY-002 — Payment shall occur via an off-platform hosted checkout; the provider shall confirm payment via an HMAC-verified webhook, processed idempotently.

FR-PAY-003 — The system shall maintain payment statuses: Pending, Completed, Failed, Refunded.

FR-PAY-004 — Users shall receive a payment-confirmation notification.

FR-PAY-005 — The reveal price shall be configurable (default 500 ETB), in ETB.

FR-PAY-006 — The system shall generate revenue reports for administrators (on-screen).

5.11 Notification Management

FR-NOT-001 — The system shall send a daily digest notifying users of new qualifying interests in their area since their last digest.

FR-NOT-002 — Administrators shall be able to send broadcast notifications to all users or a filtered segment (by bank, region, or zone).

FR-NOT-003 — The system shall send notifications for: new-match digest, payment confirmation, admin broadcasts, and profile-completion nudges.

FR-NOT-004 — Notifications shall be delivered via Telegram Bot messages only (no email/SMS fallback).

FR-NOT-005 — Notification text shall be available in both English and Amharic, rendered per the user's language preference.

5.12 Admin, Staff & Reporting

FR-ADM-001 — Administrators shall create, update, and deactivate banks, regions, zones, and grades.

FR-ADM-002 — Super Administrators shall create staff accounts and assign roles (Super Admin or Admin).

FR-ADM-003 — Administrators shall activate or deactivate staff and user accounts.

FR-ADM-004 — The Admin PWA shall provide a dashboard summary: active users, submitted interests, purchases, and revenue.

FR-ADM-005 — The system shall provide on-screen reports: user growth/activity, interest volume, purchase/revenue, and per-bank breakdowns, with date filters.

FR-ADM-006 — Administrators shall be able to search and monitor users (phone masked in list view, full detail in per-user view).

5.13 Role-Based Access Control

FR-RBAC-001 — The system shall enforce role-based permissions across the Mini App and Admin PWA.

FR-RBAC-002 — Users shall only access functions authorized for their role.

FR-RBAC-003 — Roles shall be: Super Administrator, Administrator, and Registered Employee (User).

6. Business Rules

BR-001 — Matching shall never cross bank boundaries.

BR-002 — A candidate matches the viewer when the candidate has an interest in the viewer's zone (zone-level) or the viewer's region (region-broad interest, zone_id null).

BR-003 — Grade must be within ± 1 band (bands of 3 grades) for a candidate to be a valid match.

BR-004 — A mutual match exists when the viewer also has an interest in the candidate's zone or region; mutual matches rank above one-directional matches.

BR-005 — Contact details are revealed to a buyer only after a confirmed, successful payment.

BR-006 — A purchased reveal is permanent for that buyer/candidate pair and shall never be charged twice.

BR-007 — The platform does not execute, approve, or mediate any transfer; resolution happens outside the system.

BR-008 — Feed results are generated live at request time and are not stored as persistent match records.

BR-009 — A user may select at most 3 regions and at most 3 zones per region; a zone selection implies its region; a region may be selected without zones.

BR-010 — Marketplace participation (browsing or being browsed) requires a complete profile.

BR-011 — Branch and Neighborhood are required free-text fields (at least one language version) and are paywalled (revealed only on purchase).

BR-012 — For each bilingual user field, at least one language version must exist; both may not be empty simultaneously.

BR-013 — Paywalled fields are returned as “*” from the API for unpurchased candidates; the frontend renders whatever it receives without client-side paywall logic.

7. Data Model

The detailed ERD shall be produced during system design. Core entities:

- Bank — name, name_am, nickname, is_active
- Region — name, name_am, is_active
- Zone — name, name_am, region_id (FK → Region), is_active
- Grade — grade_number, band_number, band_label/_am, tier_classification/_am, rank_order, is_active
- User — telegram_id, telegram_username, phone_number, bank_id, region_id, zone_id, branch_name_en, branch_name_am, neighborhood_en, neighborhood_am, grade_id, full_name_en, full_name_am, photo_url, photo_source, preferred_language, is_active, profile_completed_at, last_digest_at, timestamps
- TransferInterest — user_id, region_id, zone_id (nullable = broad region interest), created_at
- Purchase — buyer_id, target_user_id, revealed_fields, payment_id, timestamps
- Payment — purchase_id, provider, provider_charge_id (unique, idempotency key), amount, currency, status, raw_payload, timestamps
- Notification — user_id, type, payload, sent_at
- Staff — full_name, email, password_hash, role, preferred_language, is_active
- StaffRefreshToken — staff_id, token_hash, expires_at, revoked_at
- AuditLog — polymorphic actor_type/actor_id, action, entity_type/entity_id, metadata

Key Relationships

- One Bank has many Users.
- One Region has many Zones; one Zone belongs to one Region.
- One User has one Bank, one Region, one Zone, one Grade.
- One User has many TransferInterests (each referencing a region and optionally a zone).
- One User (buyer) makes many Purchases; each Purchase reveals one target User.
- One User receives many Notifications.

Bilingual Field Rules

Category	English	Amharic	Constraint
Reference data (banks, regions, zones, grades)	NOT NULL	NOT NULL	Both mandatory (admin-seeded)
User-entered (name, branch, neighborhood)	NULL	NULL	At least one required (CHECK constraint + app validation)

DB-Level Enforcement (user-entered bilingual fields)

```
ALTER TABLE users
```

```
  ADD CONSTRAINT chk_full_name_bilingual
```

```
  CHECK (full_name_en IS NOT NULL OR full_name_am IS NOT NULL),
```

```
  ADD CONSTRAINT chk_branch_bilingual
```

```
  CHECK (branch_name_en IS NOT NULL OR branch_name_am IS NOT NULL),
```

```
  ADD CONSTRAINT chk_neighborhood_bilingual
```

```
  CHECK (neighborhood_en IS NOT NULL OR neighborhood_am IS NOT NULL);
```

Empty strings are normalized to NULL at the application layer (trim + nullify) before insert, so the IS NOT NULL CHECK is airtight.

Note: There is no closure table. Zone-level matching is a direct equality on zone_id; region-broad matching is an equality on region_id where the interest's zone_id is null. This keeps the matching query a simple indexed lookup.

8. Integration Requirements

8.1 Telegram Platform

- Telegram Bot API for onboarding, photo fetch (getUserProfilePhotos), and notifications.
- Telegram Mini App (WebApp) for profile, marketplace, purchases, and interests, authenticated via WebApp initData validation.

8.2 Payment Provider

- Provider-agnostic interface supporting Chapa and/or Telebirr (final provider TBD).
- Off-platform hosted checkout; HMAC-verified webhook confirmation; ETB.

8.3 Caching & Background Processing

- Redis for feed caching, rate limiting, bot wizard sessions, and distributed locks.
- BullMQ (Redis-backed) for digest fan-out, broadcast fan-out, and payment-webhook post-processing, with retry/backoff.

8.4 File Storage (Local)

- Photos stored on local disk via multer diskStorage in a dedicated /avatars directory.
- Served statically (Express static middleware or nginx).
- The /avatars directory shall be included in daily backups.

- Deployment constraint: because storage is local, the API shall run as a single instance for MVP, or multiple instances shall share a persistent volume with nginx serving /avatars/* statically.

9. Security Requirements

SEC-001 — All communication shall use HTTPS/TLS.

SEC-002 — Admin passwords shall be securely hashed (bcrypt).

SEC-003 — Mini App requests shall be authenticated via Telegram WebApp initData signature verification.

SEC-004 — Role-based authorization shall be enforced on all Admin PWA and API endpoints.

SEC-005 — Failed admin login attempts shall be rate-limited (5 failures → 15-minute lockout per IP).

SEC-006 — All sensitive user, admin, and payment activity shall be recorded in audit logs; phone numbers masked in admin list views.

SEC-007 — Payment webhook endpoints shall validate HMAC authenticity and process confirmations idempotently.

SEC-008 — API endpoints shall be rate-limited (Redis-backed) to prevent feed scraping and purchase abuse.

SEC-009 — Admin sessions shall use a short-lived access token + rotating refresh token, with idle timeout.

SEC-010 — A candidate's paywalled contact information shall be returned as “*” from the API for every user except a confirmed buyer of that reveal.

SEC-011 — User-scope tokens shall be rejected on admin routes and vice versa (server-side).

SEC-012 — Photo uploads shall be validated (size cap, MIME whitelist, sanitized filenames) to prevent abuse and path traversal.

10. Telegram Bot, Mini App & Admin PWA Requirements

Telegram Bot

- Minimal onboarding: language → phone → bank → region → zone.
- Auto-fetch Telegram profile photo on registration.
- “Open Mini App” deep-link after onboarding.
- Notifications: digest, payment confirmation, broadcast, profile-completion reminder.

Telegram Mini App (4 tabs)

- Feed — ranked endless scroll of candidates who want the viewer's area; mutual-first; * masking; unlock button; completeness nudge at top; back-to-top control.
- People — candidates in the viewer's desired areas; requires interests (empty state otherwise).
- Purchases — history of unlocked contacts (fully visible).
- Profile — view/edit profile, photo upload, interest management, language; completeness nudge.

- Profile-completion gate for incomplete users.

Admin PWA

- Dashboard: users, interests, purchases, revenue.
- Reference data management: banks, regions, zones, grades.
- Staff management (Super Admin only).
- User monitoring (search, per-user detail, activate/deactivate).
- Broadcast composer (all / by bank / by region / by zone), bilingual.
- On-screen reports with date filters.

Supported Languages

- English and Amharic across bot, Mini App, and Admin PWA.

11. Non-Functional Requirements

Performance

- Feed response time under 3 seconds at 1,000+ users (target sub-200 ms with caching).
- Reveal confirmation within 5 seconds of payment confirmation.
- Payment status reflected within 1 minute (webhook-driven).

Availability

- 99% uptime target.

Scalability

- Designed for 1,000+ users from launch, using Redis caching and queue-based notification fan-out.
- Flat zone/region matching keeps queries indexed and performant as volume grows.

Maintainability

- Pure JavaScript with raw SQL (mysql2/promise); module-based code structure (auth, profile, reference, interests, feed, people, purchases, payments, notifications, admin).

Backup

- Daily MySQL backup.
- Daily backup of the /avatars photo directory.
- Daily backup of persisted Redis state.

12. User Stories

US-001 — As a bank employee, I want to register quickly through the Telegram bot so I can start with minimal friction.

US-002 — As a bank employee, I want to complete my profile in the Mini App so I can access the marketplace.

US-003 — As a bank employee, I want my Telegram photo used automatically so my card looks complete without effort.

US-004 — As a bank employee, I want to add up to three desired regions/zones so I can target my transfer.

US-005 — As a bank employee, I want mutual swaps to appear at the top of my feed so I see the best opportunities first.

US-006 — As a bank employee, I want my feed to keep showing relevant people as I scroll so it never feels empty.

US-007 — As a bank employee, I want to see who's already in my dream areas so I can reach out proactively.

US-008 — As a bank employee, I want to pay to unlock a match's contact so I can reach them directly.

US-009 — As a bank employee, I want to be notified of new matches so I don't have to keep checking.

US-010 — As an administrator, I want to manage banks, regions, zones, and grades so matching stays accurate.

US-011 — As an administrator, I want to view revenue and activity reports so I can monitor platform health.

US-012 — As an administrator, I want to broadcast bilingual notifications to segments so I can drive engagement.

13. Acceptance Criteria

Profile Completion

- Marketplace features are inaccessible until at least one language version of Full Name, Branch, and Neighborhood is set, AND Grade is selected.
- Completeness nudge appears at the top of the Feed and in Profile until 100% complete.
- Nudge is language-aware (prompts for the missing language).

Transfer Interest

- User can select up to 3 regions and up to 3 zones per region; saving works with a single zone.
- Selecting a zone auto-selects its region; a region can be saved without zones.

Feed

- Shows only same-bank, complete, active candidates within ± 1 band who want the viewer's zone or region.

- Mutual matches rank first; zone-level above region-level; grade proximity then recency.
- Feed broadens from zone to region as the user scrolls and never empties while candidates want the viewer's region.
- Paywalled fields returned as “*” until purchased; full values after purchase.
- Frontend renders whatever the API returns — no client-side paywall logic.

People Tab

- Requires interests; shows empty state otherwise.
- Shows candidates in the viewer's desired areas, ranked by grade proximity then recency.

Purchase/Reveal

- Payment confirmed via provider webhook before reveal; idempotent.
- Reveal is permanent for the buyer/target pair; buyer never charged twice.
- Revealed contact visible in both Feed and Purchases.
- API returns full field values (both language versions where available) after purchase.

Notifications

- Daily digest reaches users with new qualifying matches, as a batched job.
- Broadcasts deliver bilingual messages to the selected segment.

Profile Self-Service

- Region and zone changes are fully self-service (validated against seeded reference data, no admin approval).
- Bank is not changeable after registration.

14. Future Enhancements

- Expansion beyond banks to private organizations and government institutions.
- Full transfer-request and approval workflow (mutual acceptance, HR review, completion tracking).
- Multi-party chain transfer detection ($A \rightarrow B \rightarrow C \rightarrow A$).
- Ranked/weighted transfer preferences (1st choice, 2nd choice).
- Automated KYC / identity verification.
- Transfer-demand hotspot analytics.
- In-app secure messaging between matched parties.
- Cloud object storage for photos if horizontal scaling demands it.
- Additional languages beyond English and Amharic.

15. Success Metrics

- Registered employees across participating banks.
- Profile-completion rate (registered → complete).
- Volume of transfer interests submitted.
- Feed engagement (views, scroll depth, refreshes).
- Purchase conversion rate (feed views → paid reveals).
- Revenue generated.
- Digest open/response rate.
- Platform uptime and query performance at scale.

16. Open Items & Assumptions

Open Items

#	Item	Status
1	Final payment provider — Chapa, Telebirr, or both	Open — provider-agnostic interface supports either; decision deferred

Assumptions (confirmed)

#	Assumption
1	The reveal price defaults to 500 ETB and is configurable.
2	The Telegram profile photo is displayed to same-bank users; the user consents by registering and may replace or remove it.
3	The API runs as a single instance for MVP (local photo storage); horizontal scaling requires a shared persistent volume.
4	Amharic reference-data strings (zones/grades) require translator review before production seeding.
5	For user-entered bilingual fields, at least one language version is required at input; the other is encouraged via the completeness nudge.
6	Region/zone changes after registration are fully self-service (no admin verification), as the data is pre-seeded and validated.
7	Card layout and visual design are the frontend's responsibility; the backend's contract is to return “*” for paywalled fields (unpaid) or full values (paid).